

NeuroTune



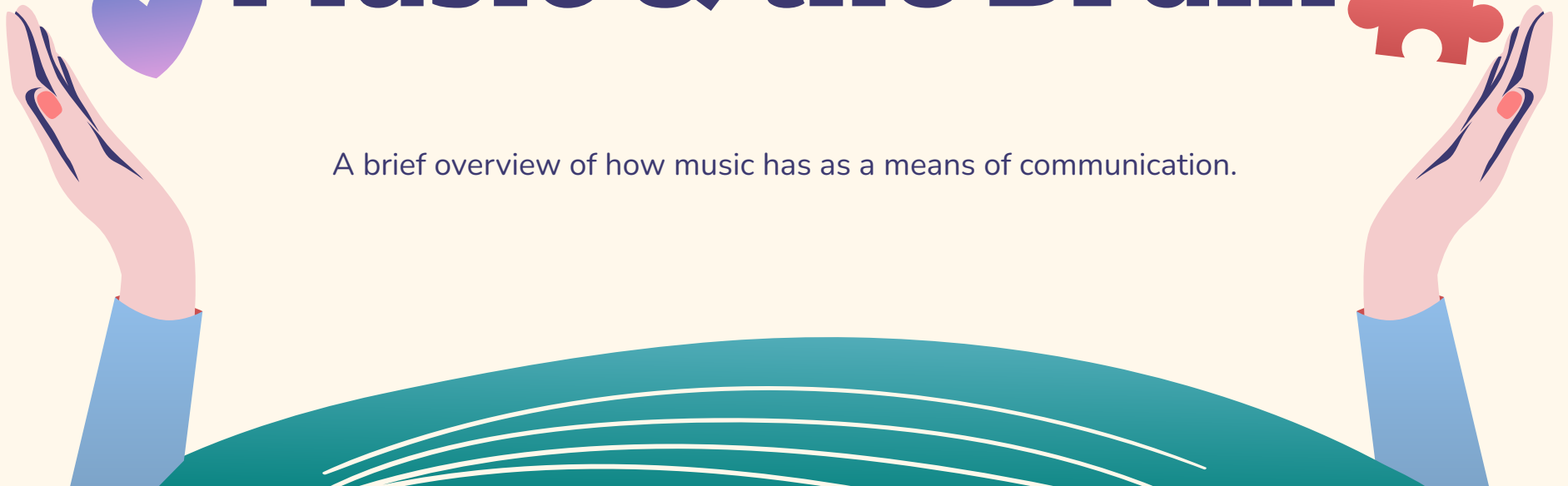
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01

Music & the Brain

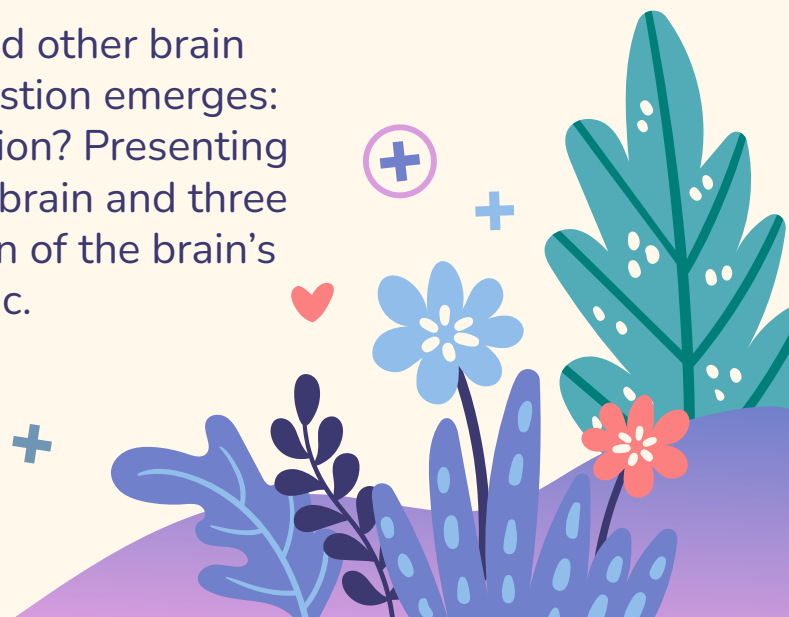


A brief overview of how music has as a means of communication.



Abstract

With the introduction of EEG (and other brain monitoring technology) a new question emerges: What are the limits of communication? Presenting Neurotune - a bridge between the brain and three LEDs. An interactive approximation of the brain's interpretation of music.



Musical Analysis

Right Hemisphere Association

- ❑ Stronger when active listening
- ❑ Pitch & melodic Aspects

Alpha Asymmetry

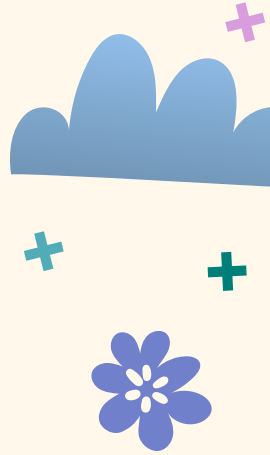
- ❑ Inverse

Left Hemisphere Association

- ❑ Temporal & Sequential (sound)
- ❑ Rhythm perception, discrimination

Approach-Withdrawal Model

- ❑ Anger
- ❑ Left vs Right





Demo!

Two pieces will be played:

- ❑ Massenet - Meditation Thaís
- ❑ Kreutzer - Etude No. 2 (Allegro Moderato)

*The lights on the light box will activate in response. green is the approximation that the left side is more active, yellow is neutral, red more indicative of the right side.



02

Setup

Methodology

Code

01

Light up system

Input &
Output

02

Connect to EEG,
utilize

Collect

03

Data output from
brain activity

Analyze

04

What does this
mean?



How The Code Works

The electro cap pulls live data from two electrodes, F3 (L) and F4 (R)

Because of static, we use a digital filter to throw everything away except for alpha waves. Which can be tied closely to relaxation and engagement

It measures how strong those alpha waves are on the left side vs right. The comparison is called frontal asymmetry

Before we start the demo, it records ten seconds of baseline brain activity as a starting benchmark

It then takes the final math score and each led is correlated with a state, Left, Right, Neutral.



Math!

- 
- 
- alpha wave strength. filtered 8-12Hz. both sides
 - calculate their power
 - squares them
 - positive and negative values both become positive.
 - average the last 3 scores
 - final score- crossed a the trigger line = changes states.

03 Ending Notes



Caveat



Equipment

EEG, Setup,
code



Bias

Likes, Dislikes,
etc.



Methods

Monitoring,
Music



Environment

Presentation, time
constraint, stress,
other input

Importance of Simplicity

- ❑ Focusing on MVP over perfection
- ❑ Prioritise strong conceptual idea
- ❑ Treating music as art and communication not rigidly mathematical

Application - Human



Neurodivergence

Therapeutic and
artistic outlet



Alternate Spoken Languages

Shared appreciation or
dislike.



Small Children

Musical understanding
even before speech.



Possible Future Study - Animals



Communication



Music Therapy



Rehabilitation



Recap



Abstract

NeuroTune



Setup

Methods + Research



Findings

Live Demo



Caveats

Constraints + other factors





Conclusion

Moving forward....



Sources

Frontal and parietal EEG alpha asymmetry: a large-scale investigation of short-term reliability on distinct EEG systems

From Neural Networks to Emotional Networks: A Systematic Review of EEG-Based Emotion Recognition in Cognitive Neuroscience and Real-World Applications

Asymmetric hemisphere activation in tenderness: evidence from EEG signals

How musical training affects cognitive development: rhythm, reward and other modulating variables

Live music stimulates the affective brain and emotionally entrains listeners in real time

Lateralization of auditory-cortex functions

The neuroscience of music perception: a narrative review

